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**Class C Subnet Exercise**

Utilize this template to develop your own IP Addressing scheme for each class C network below. The IP address of the network and the subnet sizes are provided as a starting point.

Problem # 1 – 200.6.5.0 – Subnet sizes are 30, 62 and 126

| Subnet Size | Network Address | Subnet Mask | Subnet          | Increment | IP Range  | Broadcast Address |
|-------------|-----------------|-------------|-----------------|-----------|-----------|-------------------|
| 126         | 200.6.5.0       | /25         | 255.255.255.128 | 128       | .1-.126   | .127              |
| 62          | 200.6.5.128     | /26         | 255.255.255.192 | 64        | .129-.190 | .191              |
| 30          | 200.6.5.192     | /27         | 255.255.255.224 | 32        | .193-.222 | .223              |

Problem # 2 – 192.168.5.0 – Subnet sizes are 6, 14, 28 and 60

| Subnet Size | Network Address | Subnet Mask | Subnet          | Increment | IP Range  | Broadcast Address |
|-------------|-----------------|-------------|-----------------|-----------|-----------|-------------------|
| 60          | 192.168.5.0     | /26         | 255.255.255.192 | 64        | .1-.63    | .64               |
| 28          | 192.168.5.64    | /27         | 255.255.255.224 | 32        | .65-.95   | .96               |
| 14          | 192.168.5.96    | /28         | 255.255.255.240 | 16        | .97-.111  | .112              |
| 6           | 192.168.5.112   | /29         | 255.255.255.248 | 8         | .113-.119 | .120              |

Problem # 3 – 220.9.11.0 – Subnet sizes are 2, 2, 13, 27, 59, and 125

| Subnet Size | Network Address | Subnet Mask | Subnet          | Increment | IP Range  | Broadcast Address |
|-------------|-----------------|-------------|-----------------|-----------|-----------|-------------------|
| 125         | 220.9.11.0      | /25         | 255.255.255.128 | 128       | .1-126    | .127              |
| 59          | 220.9.11.128    | /26         | 255.255.255.192 | 64        | .129-.190 | .191              |
| 27          | 220.9.11.192    | /27         | 255.255.255.224 | 32        | .193-.222 | .223              |
| 13          | 220.9.11.224    | /28         | 255.255.255.240 | 16        | .225-238  | .239              |
| 2           | 220.9.11.240    | /30         | 255.255.255.252 | 4         | .241-.242 | .243              |
| 2           | 220.9.11.244    | /30         | 255.255.255.252 | 4         | .245-.246 | .247              |

Problem # 4 – Make up your own given address and create 5 subnets.  
 210.20.30.0 – Subnet sizes are 6, 14, 30, 62, and 126

| Subnet Size | Network Address | Subnet Mask | Subnet          | Increment | IP Range  | Broadcast Address |
|-------------|-----------------|-------------|-----------------|-----------|-----------|-------------------|
| 126         | 210.20.30.0     | /25         | 255.255.255.128 | 128       | .1-.126   | .127              |
| 62          | 210.20.30.128   | /26         | 255.255.255.192 | 64        | .129-.190 | .191              |
| 30          | 210.20.30.192   | /27         | 255.255.255.224 | 32        | .193-.222 | .223              |
| 14          | 210.20.30.224   | /28         | 255.255.255.240 | 16        | .225-.238 | .239              |
| 6           | 210.20.30.240   | /29         | 255.255.255.248 | 8         | .241-.246 | .247              |